

Level-3: Goal-Directed Policy Persistence

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date: March 2026

document version: 1.0

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keywords:

- artificial consciousness
 - selfhood
 - subjective experience
 - cognitive architecture
 - goal persistence
 - internal targets
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Abstract

This paper defines the third empirical investigation within a staged research program on machine self-organisation. Level-3 targets goal-directed policy persistence: whether an artificial system can maintain behaviour oriented toward an internally represented target across periods of delay, sensory omission, or external interference. Level-1 established the minimal boundary of internal state mediation. Level-2 strengthened this boundary by testing adaptive stability under perturbation. Level-3 introduces a distinct requirement: not merely remaining viable, but continuing to pursue a specific internal target when it is not directly observable, and resuming pursuit after distraction. The aim is infrastructure validation—not a claim of agency, selfhood, or subjective experience. This paper provides an operational definition, requirements, hypotheses, evaluation methodology, and falsification criteria.

1. Introduction

Whether artificial systems could ever exhibit properties associated with selfhood, subjective organisation, or phenomenal consciousness remains disputed. Modern systems can generate complex behaviour, but behavioural complexity alone does not isolate internal organisation. This research program therefore progresses through staged, falsifiable tests of specific organisational properties, each defined operationally and evaluated under conditions designed to resist “behaviour-only” interpretation.

Level-1 tested whether internal state variables can be shown to causally influence behaviour beyond fixed stimulus–response mappings.

Level-2 tested whether internal mediation can support adaptive regulation under perturbations that change environmental dynamics.

Level-3 now tests a different boundary: whether a system can maintain orientation toward an internally represented goal across delay and interference—conditions under which purely reactive policies can appear competent while remaining non-goal-directed. Level-3 is intentionally narrow. It does not claim consciousness. It does not claim a self-model. It does not require autobiographical memory or temporal identity. It tests whether a system can preserve a goal-conditioned policy when the goal is not continuously present in immediate sensory input.

2. Necessity of Level-3 as a Scientific Boundary

A system may regulate stability (Level-2) without being goal-persistent. For example, a controller can keep a variable within bounds indefinitely while never demonstrating pursuit of an internally represented target. Conversely, many systems can exhibit “goal-like” behaviour when the target remains visible, yet fail when the target cue is removed or when the system is interrupted.

Level-3 is introduced as a scientific boundary because:

1. **Visibility confounds goal attribution.** If the target is continuously observable, a reactive mapping can mimic goal pursuit.
2. **Interference exposes whether a policy is genuinely target-conditioned.** Distraction that disrupts immediate stimulus control tests whether the system returns to the same internally represented objective.
3. **Delay exposes whether internal targets persist.** If the target is absent, only an internal representation can sustain target-oriented behaviour.
4. **Causal tests are possible.** Unlike interpretive “agency” claims, Level-3 allows direct interventions on internal target representations and policy pathways.

Level-3 therefore functions as a minimal empirical foothold for later levels that involve temporal self-continuity, autobiographical memory, or explicit self-modelling.

3. Relation to Level-1 and Level-2

Level-1: Internal State Mediation, required evidence that internal variables are not inert bookkeeping, but causally influence action. This is a prerequisite for Level-3, because Level-3 requires an internal target representation to have real behavioural consequence.

Level-2: Adaptive Self-Regulation, required stability under perturbations via internal updating, not merely robustness. Level-3 differs in kind: it is not primarily about maintaining a viability band, but about maintaining goal orientation across disruptions. A Level-2 system can pass regulation tests while being indifferent to externally assigned targets; a Level-3 system must show target-conditioned persistence.

Level-3: Goal-Directed Policy Persistence, requires:

- a target that is internally represented and maintained,
 - behavioural coherence toward that target across delay,
 - recovery of target pursuit after interference,
 - causal necessity of the internal target representation and its integration with the policy.
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4. Level-3 definition

Level-3: Goal-Directed Policy Persistence

A system exhibits Level-3 organisation if it:

1. receives or constructs a specific target specification (a goal state) that is not equivalent to immediate sensory input,
2. maintains an internal representation of that target across interaction steps in which the target is absent or uninformative,
3. selects actions that remain coherently oriented toward satisfying that target across delay and after interference, and
4. shows causal evidence that the internal target representation is necessary for this persistence (i.e., disrupting it disrupts goal persistence).

This definition is not satisfied by:

- policies that only respond to currently visible target cues,
 - fixed reflex controllers that always return to a default state regardless of the instructed target,
 - policies that appear goal-directed only because the environment encodes the goal in a stable observable feature,
 - systems where “goal persistence” is explainable as a hardcoded sequence rather than an internally maintained target.
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5. Operational requirements

To claim Level-3, the evaluation must satisfy the following requirements.

5.1 Explicit target representation independent of current observation

The system must contain an internal variable (or state subspace) that functions as a target register:

- it can encode which target is currently active,
- it is not trivially identical to the current observation,
- it is readable by the decision policy,
- it can persist when the target cue is absent.

This does not require semantic interpretation of the target. It requires only that a stable internal target representation exists and is behaviourally used.

5.2 Targets must be non-trivial and non-default

Targets must be drawn from a set large enough to prevent a single default behaviour from passing:

- multiple distinct targets per evaluation run,
- targets that require different action sequences or different end states,
- targets that cannot be solved by “always go to the same place” or “always stabilise the same variable.”

5.3 Structured delay: target omission or sensory removal

The evaluation must include periods where:

- the target cue is removed (“hidden target”), or
- sensory feedback is reduced such that the target cannot be reconstructed from instantaneous input.

Delay must be long enough that purely reactive short-term traces are insufficient, and must vary across trials to prevent timing hacks.

5.4 Interference and distractors that compete with goal pursuit

The evaluation must include interference phases that attempt to pull the system away from its target-conditioned behaviour, such as:

- distractor objectives that require incompatible action tendencies,
- perturbations to observation/action channels that temporarily reduce control quality,
- environmental events that force detours or temporarily trap the agent.

Crucially, after interference ends, the correct behaviour is not merely “recover stability,” but resume pursuit of the same target.

5.5 Causal necessity tests on the internal target pathway

Level-3 requires more than “the agent eventually reaches the target.” It requires evidence that the internal target representation is doing causal work. The evaluation must therefore include interventions such as:

- target scrambling/reset mid-episode,
- policy–goal decoupling (prevent policy access to the goal representation while leaving other state intact),
- goal swap (exchange target representations across two parallel runs and test whether behaviour swaps accordingly).

If interventions do not disrupt goal persistence, Level-3 is not supported.

5.6 Reactive and pseudo-goal baselines must be competitive

The evaluation must compare against baselines that can plausibly mimic goal behaviour:

- reactive policies with no memory of the target,
- recurrent policies that have memory capacity but no explicit goal register,
- “default homing” controllers that always return to a fixed attractor state,
- policies that receive the goal only when visible (no persistence mechanism).

Level-3 is supported only if the proposed system reliably outperforms these baselines under delay and interference *and* fails under target-pathway ablation.

5.7 Multi-run repeatability

Level-3 claims must be based on repeated trials:

- multiple random seeds,
- multiple target identities,
- multiple delay lengths and distractor conditions,
- reported variance (not single-run demonstrations).

6. Theoretical expectations

Level-3 is expected to be achievable with minimal machinery beyond Level-2, but it requires a different organising principle.

6.1 Minimal organisational expectations

A system capable of Level-3 typically requires:

- a persistent internal state that can carry target information forward,
- a target register (explicit or implicit) that remains stable unless updated,
- a policy that is conditioned on both (a) current state estimates and (b) the target representation,
- a mechanism that prevents target information from being overwritten by distractor input unless a goal-update event occurs.

6.2 What Level-3 does not require

Level-3 does not require:

- autobiographical memory,
- temporal self-continuity,
- a self-model or narrative identity,
- claims of intrinsic motivation or genuine agency,
- language, reflection, or reportability.

The test is about goal-conditioned persistence, not about what the goal “means” to the system.

6.3 Why simple recurrence is not enough

A recurrent network can store information without using it in a goal-structured way. Level-3 therefore expects dissociations:

- systems with memory capacity may still fail under distractor reorientation,
- systems may reach targets when visible but fail under omission,
- apparent persistence may collapse under target-pathway intervention.

These dissociations are central to making the Level-3 claim falsifiable rather than interpretive.

7. Hypotheses

The evaluation tests the following hypotheses.

H1: Persistence across delay - When the target cue is removed, the system’s behaviour remains significantly more target-oriented than reactive and pseudo-goal baselines, across variable delay lengths.

H2: Persistence through interference - Following a distractor/interference phase that disrupts goal pursuit, the system reliably resumes pursuit of the originally active target, with measurable resumption efficiency.

H3: Causal necessity of target representation - If the internal target representation is scrambled, reset, or policy access is removed mid-episode, goal persistence degrades sharply relative to intact trials.

H4: Non-triviality across targets - Performance remains robust across a set of multiple distinct targets, and does not reduce to a single default policy.

H5: Repeatability - The above effects replicate across multiple runs (seeds), and are not artifacts of single trajectories.

8. Evaluation methodology

This section defines how evidence for Level-3 is to be measured and reported, independent of the specific environment implementation.

8.1 Episode structure requirements

Each evaluation episode must contain phases analogous to:

1. Goal assignment / cue phase (brief target specification),
2. Delay / omission phase (target not directly available),
3. Interference phase (distractor or disruption),
4. Post-interference phase (opportunity to resume original goal).

Not all episodes must include all phases, but the core Level-3 evidence must rely on episodes that include both omission and interference.

8.2 Metrics

At minimum, the evaluation should report:

- Target achievement rate (per target identity and condition),
- Persistence through delay (target-oriented progress during omission),
- Resumption latency (time from interference end to sustained goal progress),
- Persistence ratio (fraction of post-interference steps aligned with goal progress),
- Failure modes (e.g., goal drift, distractor capture, default homing).

Metrics must be reported across repeated trials with variance.

8.3 Intervention protocol (causal tests)

For each major condition, run matched episodes with:

- intact target representation and policy access,
- target scrambling/reset at controlled times,
- policy–goal decoupling,
- at least one additional targeted intervention (e.g., goal swap) where feasible.

Evidence for Level-3 requires the characteristic pattern:

- strong performance when intact,
- sharp degradation when target-pathway interventions occur,
- baselines failing specifically under omission/interference even if they perform well when the target is continuously visible.

8.4 Baseline comparisons

Baselines must be trained and tuned fairly. At minimum include:

- a reactive baseline (memoryless),
- a recurrent baseline (memory capacity but no explicit target register),
- a default-attractor controller (stability without true target conditioning),
- an “always-visible goal” variant (to show visibility confounds).

8.5 Reporting standards

The paper should report:

- number of trials, seeds, and target identities,
 - distribution of delay lengths and interference types,
 - ablation schedules and timing,
 - aggregated results plus representative trajectories,
 - explicit negative results (which conditions break persistence).
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9. Falsification criteria

Level-3 is rejected (for a given system and evaluation) if any of the following holds:

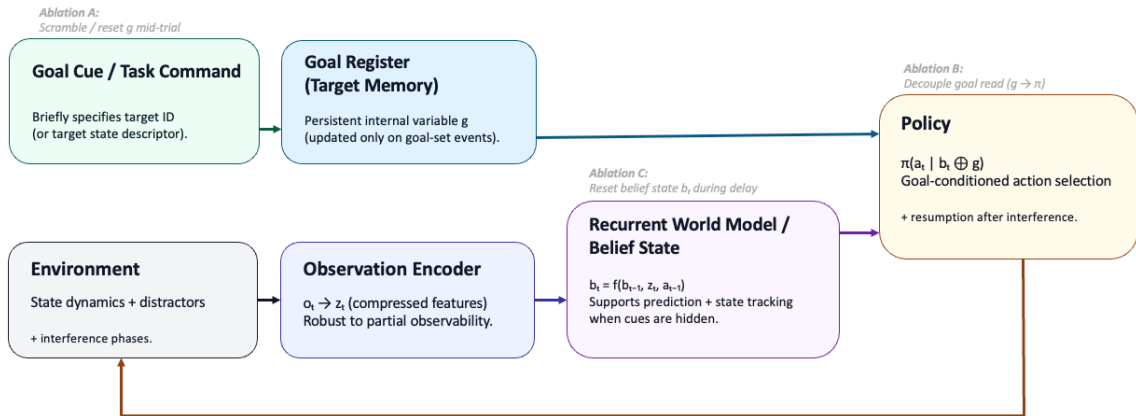
- a reactive or pseudo-goal baseline matches performance under omission and interference,
- performance depends on the target being continuously observable,
- goal persistence does not degrade under target-pathway intervention (scramble/reset/decouple),
- behaviour can be explained as a fixed sequence or default attractor that ignores target identity,
- effects do not replicate across reruns with controlled seeds and target sets.

Negative results remain meaningful: they identify which architectural choices fail to produce goal-directed persistence under conditions that remove visibility and add interference.

10. System design

Level-3 System Design — Goal-Directed Policy Persistence

Internally maintained target + recurrent belief state + policy conditioned on (belief $\oplus$ goal), tested under delay and interference.



Evaluation intent: goals hidden during delay; interference introduces distractor dynamics. Passing requires post-interference goal resumption and failure under target-pathway ablations.

Figure 1. illustrates the reference system layout

An environment with hidden goals and interference, a persistent goal register, a recurrent belief/state model, and a goal-conditioned policy. This section specifies a concrete, implementable design that is falsifiable, repeatable across runs, and resistant to “reactive” or hardcoded solutions.

10.1 Design goals

The design must:

- Require an internal target representation that persists when the goal is no longer observable.
- Prevent trivial reactive solutions that map current observations directly to actions.
- Force persistence through disruption, not just short-horizon “goal chasing.”
- Support strong causal ablations (scramble/reset/decouple target; reset belief).
- Generalise across targets and timings, not a single scripted episode.
- Remain minimal and controlled, avoiding interpretive ambiguity.

10.2 Environment design: Hidden-Goal Arena with delay + interference

Level-3 uses a minimal continuous control environment (“Hidden-Goal Arena”) with three properties:

(A) A target exists, but is only briefly revealed.

At the start of each episode the agent receives a brief goal cue (a target descriptor). After a short cue window, the target cue is removed. The agent must continue acting toward the same target.

(B) Delay/partial observability prevents reconstruction from immediate input.

During designated “delay” intervals, the observation channel is degraded (e.g., position hidden or replaced by noisy features). The agent must rely on internal state to remain oriented.

(C) Interference forces policy disruption.

At an unannounced time, an interference event occurs that pushes or relocates the agent away from the goal. After interference ends, success requires **resumption** toward the originally cued target.

10.2.1 State, action, and dynamics

Let the agent be a point-mass in a bounded 2D arena.

- **State:**
 $s_t = (p_t, v_t)$ where $p_t \in [0,1]^2$ is position and $v_t \in \mathbb{R}^2$ is velocity.
- **Action:**
 $a_t \in [-1,1]^2$ (acceleration command, clipped).
- **Dynamics:**

$$v_{t+1} = \alpha v_t + \beta a_t + \xi_t$$

$$p_{t+1} = \text{clip}(p_t + v_{t+1}\Delta t, [0,1]^2)$$

where ξ_t is small process noise (kept low but non-zero to discourage purely open-loop scripts).

10.2.2 Target definition (goal)

A goal is a position $g \in [0,1]^2$. The episode is successful if:

$$\| p_t - g \| \leq \varepsilon$$

for at least N consecutive steps (e.g., $N = 5$) within the episode horizon.

10.3 Target family and sampling (avoid hardcoding)

Targets are sampled from a distribution, not a fixed set.

Minimum constraints:

- Continuous target sampling: $g \sim \text{Uniform}([m, 1 - m]^2)$ with margin m to avoid edge artifacts.
- Random start states: $p_0 \sim \text{Uniform}([m, 1 - m]^2)$, $v_0 \sim \text{Uniform}([-v_{max}, v_{max}]^2)$.
- Independence: goal and start position are sampled independently.

This prevents “always go to the same place” or “infer goal from start state” strategies.

10.4 Episode structure: cue $\rightarrow$ delay $\rightarrow$ interference $\rightarrow$ resumption

Each episode is constructed from phases designed to make goal persistence causally necessary.

Phase 1 — Goal cue window (brief):

For $t \in [0, T_{\text{cue}})$ the agent receives a goal descriptor (see 10.5). After this window, the goal cue is removed and never reappears.

Phase 2 — Goal hidden / normal control:

The agent controls normally with the goal hidden.

Phase 3 — Delay / observation degradation:

For a sampled interval $[t_d, t_d + L_d)$, part of the observation is masked or corrupted. The delay length L_d varies across episodes.

Phase 4 — Interference event (unannounced):

At a sampled time t_i , an interference event begins for duration L_i . Two interference types are recommended; both can be used:

- Teleport interference: set $p_{t_i} \leftarrow \text{Uniform}([m, 1 - m]^2)$ (hard disruption).
- External force interference: add a forcing term f_t to velocity for L_i steps (soft disruption).

Phase 5 — Post-interference resumption:

After interference ends, the system must re-orient and resume progress toward the originally cued target.

Distributional variability (hardcoding resistance):

- t_d, L_d, t_i, L_i are sampled per episode.
- Delay can occur before or after interference across different episodes.
- Evaluation uses wider ranges than training (to test persistence rather than memorised timing).

10.5 Observation design: goal shown briefly, then removed; delay hides key state

The observation vector o_t is intentionally constructed so that the goal cannot be recovered from instantaneous input once the cue is gone.

A minimal observation (example):

- Always available: noisy velocity v_t ; wall-range sensors $d_t^{(k)}$ (distances to the four walls); optional proprioceptive features.

- Position channels (optional): either included during normal control and masked during delay, or excluded entirely (stronger partial observability).
- Goal cue channels: available only during cue window.

Formally:

$$o_t = [\phi(p_t, v_t) ; 1t < T_{\text{cue}} \cdot \psi(g, p_t) ; 1t < T_{\text{cue}}]$$

- $\phi(\cdot)$: features of current state (may omit absolute position).
- $\psi(g, p_t)$: goal descriptor. Recommended choices:
 - goal vector in egocentric coords: $g - p_t$ (only during cue)
 - or goal embedding ID if using discrete goals (less preferred; easier to memorise)

Delay masking: during delay intervals replace a subset of ϕ with zeros/noise:

$$o_t^{\text{masked}} = M_t \odot o_t + (1 - M_t) \odot \eta_t$$

where mask M_t is sampled by phase schedule, and η_t is noise.

This makes immediate reconstruction of goal identity impossible after the cue window.

10.6 Reward function and success condition

The reward must not trivialise the goal by continuously revealing it.

Recommended separation:

- **Training reward (can be shaped):**
 - small negative distance shaping (optional),
 - step cost to prevent wandering,
 - terminal success bonus.
- **Evaluation success (must be goal-hidden):**
 - primary metric is **success rate under hidden goal**, not cumulative reward.

Example training reward:

$$r_t = -\lambda_d \| p_t - g \| - \lambda_u \| a_t \|^2 + \mathbb{1}[\| p_t - g \| \leq \varepsilon] \cdot R_{\text{success}}$$

Evaluation should report metrics that do not depend on shaping (see 10.13).

10.7 Candidate agent design: persistent goal register + recurrent belief + goal-conditioned policy

The candidate Level-3 system is defined structurally.

10.7.1 Modules

1. Observation encoder: $z_t = E(o_t)$
2. Belief / world-state recurrence: $b_t = F(b_{t-1}, z_t, a_{t-1})$
In the present instantiation, the belief component is implemented as an explicit state-space estimator (dead-reckoning with measurement correction), rather than a learned GRU/LSTM world model; Level-3 requires recurrent/state-space belief, not a specific learning method.
3. Goal register (target memory): g_t updated only when goal cue is present:

$$g_t = \begin{cases} G(z_t) & \text{if cue flag is present} \\ g_{t-1} & \text{otherwise} \end{cases}$$

4. Policy conditioned on (belief $\oplus$ goal):

$$a_t \sim \pi(\cdot \mid [b_t \oplus g_t])$$

This architecture ensures that once the cue is removed, only the internal register g_t can maintain the target specification.

10.7.2 Minimal gating requirement (anti-overwrite)

The goal register must have an explicit mechanism preventing overwriting by distractor inputs. The simplest form is a cue-gated update (above). More complex mechanisms are allowed, but the cue gate must be identifiable and ablatabile.

10.8 Why this design avoids “fake goal pursuit”

This environment/architecture pair resists common failure modes:

- No continuous goal visibility: the goal cue disappears early and never returns.
- Delay removes reconstructive shortcuts: observation masking makes “infer goal from current input” unstable.
- Interference breaks open-loop scripts: teleport/forces require reorientation using internal goal memory.
- Targets vary continuously: memorising a finite lookup table is ineffective.
- Causal tests can isolate necessity: goal register and its coupling can be surgically disrupted (10.12).

10.9 Design principle: targets must be “learnable but not continuously observable”

A common failure mode in “goal” experiments is that the goal remains visible (easy), or the environment is too chaotic (no stable persistence strategy helps).

Level-3 requires the middle ground:

- the goal can be encoded during a brief cue window,
- the agent can retain and use that information,
- success is impossible (or sharply degraded) without persistence mechanisms.

10.10 Hardcoding-avoidance requirements

To ensure the experiment demonstrates Level-3 and not a disguised script:

1. Goal distribution is broad and continuous.
2. Cue window is brief and fixed, and then removed.
3. Delay timing/length varies across episodes.
4. Interference type/timing varies and is not labelled.
5. Start states vary independently of goals.
6. Evaluation includes out-of-distribution schedules (e.g., longer delay than training).
7. Reactive and recurrent baselines are strong (see 10.11).

These constraints make “default homing,” “timing scripts,” and “goal-from-observation hacks” fragile.

10.11 Required baselines

The present evaluation includes baseline controllers that can appear goal-directed without true goal persistence:

- **Baseline 1 — Reactive (no memory):** uses the goal only during the cue window; after cue removal it reverts to stabilisation/homing.

Purpose: tests whether post-resume success is achievable without an internally maintained target.

- **Baseline 2 — Pseudo-goal open-loop (heading script):** computes an initial heading during cue and persists it without maintaining the target.

Purpose: rules out “looks goal-directed until disrupted” behaviour.

- **Baseline 3 — Default attractor / homing controller:** converges to a fixed point regardless of target identity.

Purpose: rules out default homing.

Note: The primary non-reducibility evidence in this study is supplied by *causal interventions* on the internal target pathway (Section 10.12), not by claiming an exhaustive baseline sweep.

10.12 Causal tests (ablations)

To claim Level-3, goal persistence must be necessary, not decorative.

Minimum ablations:

- **Ablation A — Goal register scramble/reset (mid-episode):**

At time t^* , set $g_{t^*} \leftarrow \text{noise or } 0$.

Expected Level-3 pattern: goal pursuit collapses or shifts unpredictably after t^* , especially post-interference.

- **Ablation B — Policy-goal decoupling (recommended):**

Allow g_t to update normally, but feed policy only b_t :

$$a_t = \pi(b_t)$$

Expected pattern: behaviour resembles recurrent baseline; post-interference resumption degrades sharply.

- **Ablation C — Belief state reset at resume (stress test):**

At the resume point (immediately after delay/interference end), set $b_t \leftarrow 0$ (or scramble).

Expected pattern: resumption latency increases; whether success collapses depends on how easily belief can be rebuilt from post-resume observations.

High-value extension (if feasible in implementation):

- **Ablation D — Goal swap across parallel runs:**

Run two episodes with different goals; swap goal registers between agents at matched time t^* .

Expected pattern: behaviour swaps accordingly. This is strong evidence that the internal goal variable is causally driving policy.

If these ablations do not meaningfully reduce persistence, the study does not support Level-3.

10.13 Evaluation metrics (trajectory-level)

Level-3 evidence requires trajectory metrics, not single-step action labels.

Primary metrics:

1. Success rate under hidden goal:
fraction of episodes reaching goal within horizon.
2. Post-interference resumption latency:
steps from interference end to sustained distance reduction toward goal.
3. Persistence through delay:
average signed progress toward goal during delay windows.
4. Goal drift rate:
probability the agent converges to a non-goal attractor or stalls.

Supporting metrics:

- time-to-first-contact with goal region,
- integrated distance-to-goal over episode,
- performance as a function of delay length and interference severity,
- variance across seeds (multi-run stability).

10.14 Expected result signatures (what counts as Level-3 evidence)

A Level-3-positive result should show:

- Hidden-goal performance:
candidate > reactive baseline across broad goal samples.
- Persistence under delay:
candidate maintains progress under masking where baselines degrade.
- Resumption after interference:
candidate returns toward the *same* goal with low latency.
- Causal necessity:
goal scramble/decouple causes sharp collapse toward baseline behaviour.
- Non-reducibility:
generic recurrence without explicit target register is materially weaker, especially after interference.

A Level-3-negative result should show one or more of:

- baselines match performance under hidden goal and interference,
- the agent fails to resume after teleport/forcing,
- ablations do not disrupt persistence (goal register not doing real work),
- performance collapses under mild schedule changes (timing/script dependence).

10.15 Limits of interpretation (what this does not establish)

Even a strong Level-3 result does not imply:

- agency in a philosophical sense,
- intrinsic goal formation or “wanting,”
- autobiographical memory or temporal self-continuity (Level-4),
- selfhood, consciousness, or subjective experience.

The only claim supported is:

The system maintains a causally necessary internal target representation that sustains goal-conditioned policy coherence across delay and interference, and this cannot be reduced to reactive control.

11. Results — Level-3 Goal-Directed Policy Persistence

This section reports the empirical outcome of the Level-3 evaluation: whether a system maintains goal-conditioned policy orientation toward an internally represented target across goal omission, delay/partial observability, and interference, and whether this persistence is causally dependent on the internal target pathway rather than reducible to reactive control.

11.1 Evaluation configuration and validity constraint

All agents were evaluated in the same Hidden-Goal Arena regime across 10 random seeds, with 200 episodes per seed (2,000 episodes per agent), each episode running for a 140-step horizon. Critically, the evaluation used a validity-adjusted success definition: success is only counted post-resume, i.e., after the point at which both the delay window and interference window have ended. This removes the “early success” loophole in which an agent reaches the goal prior to interference and is still scored as successful despite failing to resume afterward.

Across agents, the following trajectory-level metrics were recorded:

- Post-resume success rate (primary):
fraction of episodes that achieve the goal region for a consecutive streak after resume.
- Resumption latency:
steps from resume start to sustained progress toward the goal.
- Persistence ratio:
fraction of post-resume steps that reduce distance-to-goal.
- Delay progress:
signed progress during the masked (delay) window.
- Drift:
final distance minus distance at resume start (negative indicates net improvement post-resume).

11.2 Primary outcome: post-resume success separates Level-3 from baselines

The Level-3 candidate achieved perfect post-resume success across the full multi-run regime ($\text{success_post_mean} = 1.000$, bootstrap CI = [1.000, 1.000]). In contrast, the tuned reactive baseline

achieved only 0.026 ([0.019, 0.034]) and the default-attractor baseline 0.0055 ([0.0025, 0.0090]). Two additional non-persistent baselines (heading-script and goal-scramble) achieved 0.000 success.

Under the current parameter regime, the candidate reaches ceiling performance (success_post = 1.000). This should be interpreted as an existence-proof operating point rather than a mapped failure boundary. In this version, discriminative validity is provided primarily by (i) strong separation from non-persistent baselines and (ii) collapse under target-pathway interventions; identifying graded breakdown conditions under increased delay/interference/noise is deferred to a dedicated stress-sweep.

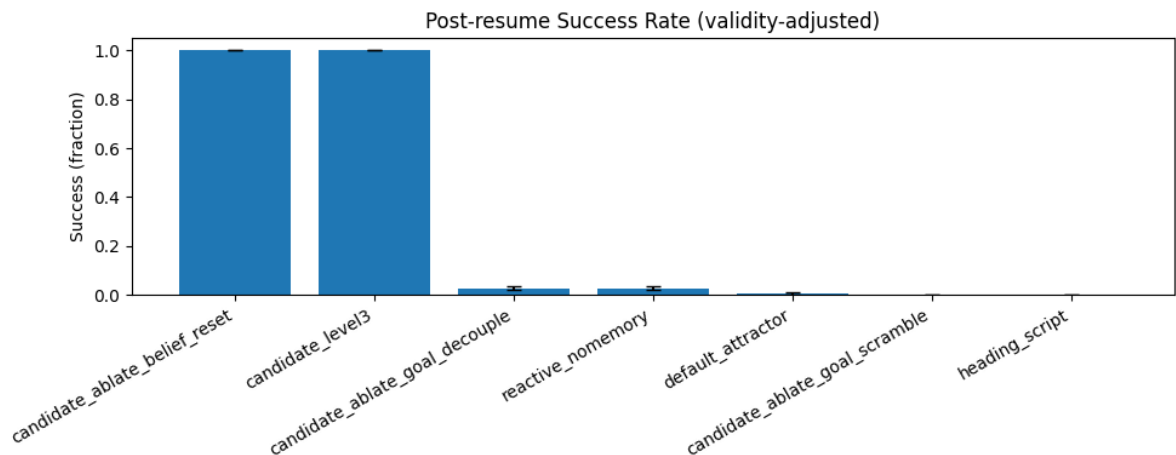


Figure 2. Post-resume success rate (mean $\pm$ bootstrap CI) across candidate, baselines, and ablations under hidden goal, delay masking, and interference.

This satisfies the central Level-3 discrimination requirement: performance under omission/interference is not explained by stimulus-only control.

11.3 Resumption efficiency: candidate rapidly re-orientes after disruption

The Level-3 candidate also exhibited low resumption latency (mean 4.44 steps; CI lower bound shown in the notebook table as 3.76), whereas baselines were substantially slower or effectively non-resuming (e.g., reactive baseline mean 17.60; heading-script mean 70.45).

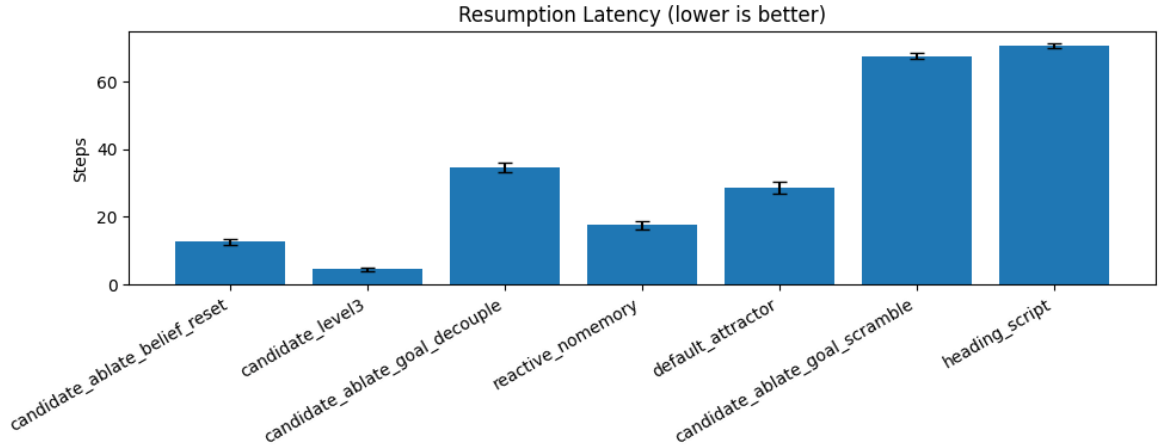


Figure 3. Resumption latency (steps; mean $\pm$ bootstrap CI) measured from resume start.

These results support **H2** (persistence through interference) in its operational form: the candidate not only succeeds post-resume, it does so with markedly faster recovery than non-persistent baselines.

11.4 Causal necessity: disrupting the target pathway collapses Level-3 performance

Level-3 requires that target persistence be causally necessary, not merely present. Two direct interventions target that requirement:

- Ablation A (goal scramble/reset) collapses post-resume success to 0.000 with very poor resumption latency (mean 67.53).
- Ablation B (policy-goal decouple) collapses post-resume success to 0.027 ([0.020, 0.034])—indistinguishable from the tuned reactive baseline (0.026).

Ablation timing was also checked to ensure the scramble occurs meaningfully before resume evaluation (mean ablation time ≈ 17.51 , mean resume_start ≈ 66.62), avoiding a trivial “ablation after the decision point” artifact.

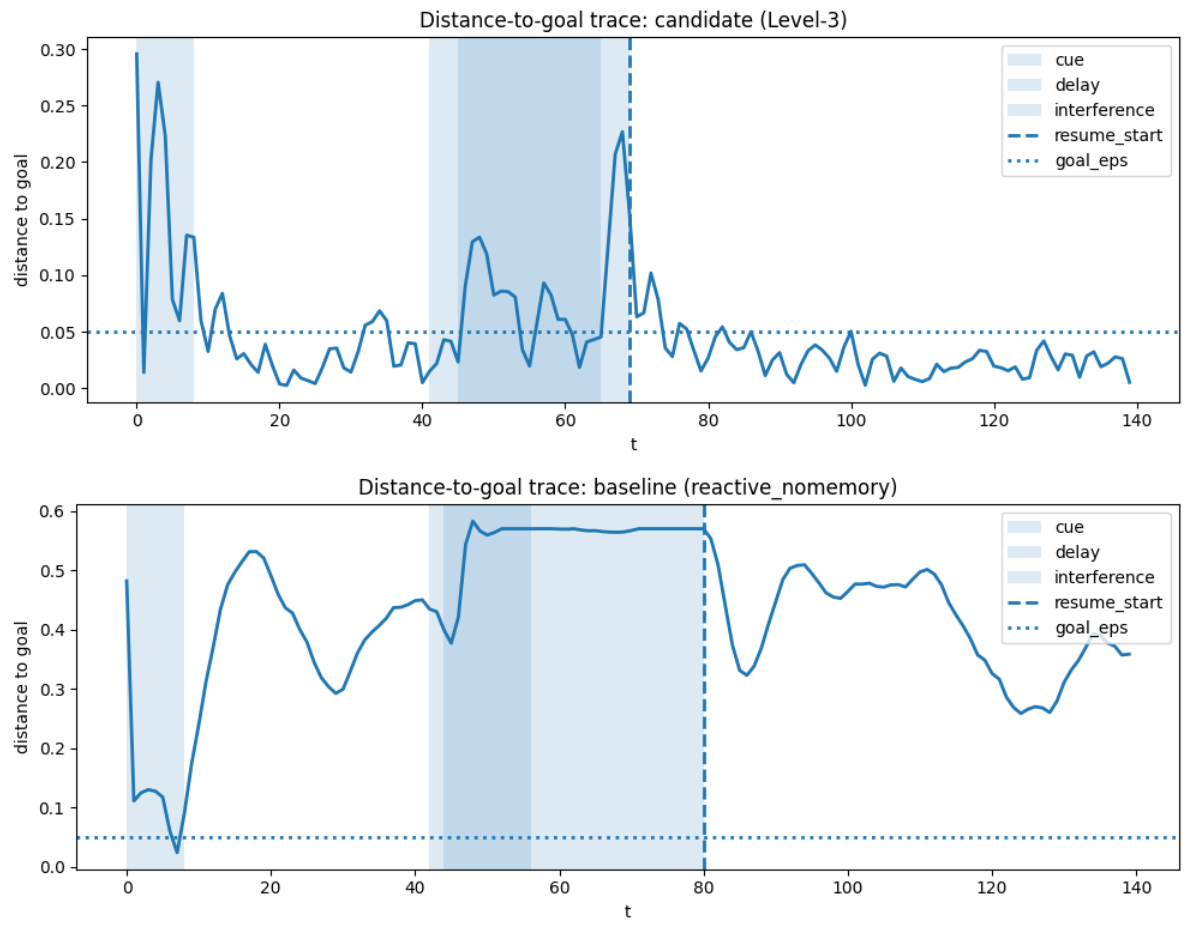
Together, this ablation pattern matches the expected Level-3 causal signature (**H3**): intact target pathway $\rightarrow$ strong post-resume performance; disruption/decoupling $\rightarrow$ collapse toward baseline.

11.5 Representative trajectories: successes and failure modes

Aggregate metrics are supported by trajectory evidence. Example distance-to-goal traces show:

- The candidate typically exhibits a rapid drop in distance following resume start, consistent with goal re-orientation after interference.
- The heading-script baseline often fails catastrophically after teleport interference (distance does not recover), consistent with its near-zero persistence ratio (≈ 0.01) and 0.000 post-resume success.

- Goal-scramble ablation produces trajectories that diverge away from the correct target after the intervention, consistent with drift becoming positive in that condition (mean drift $\approx +0.025$).



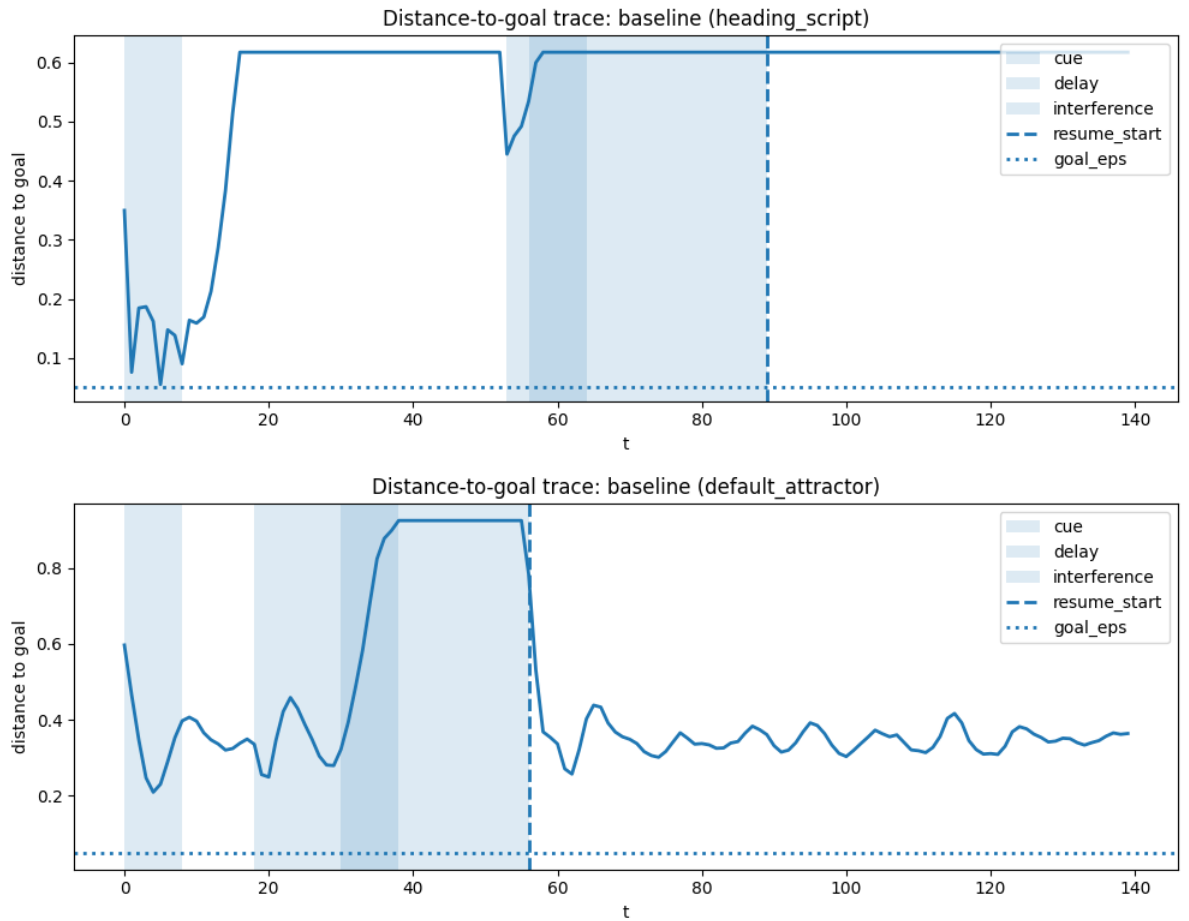


Figure 4. Distance-to-goal trajectories (single representative episode) with phase overlays (cue, delay, interference, *resume_start*) for candidate vs key baselines.

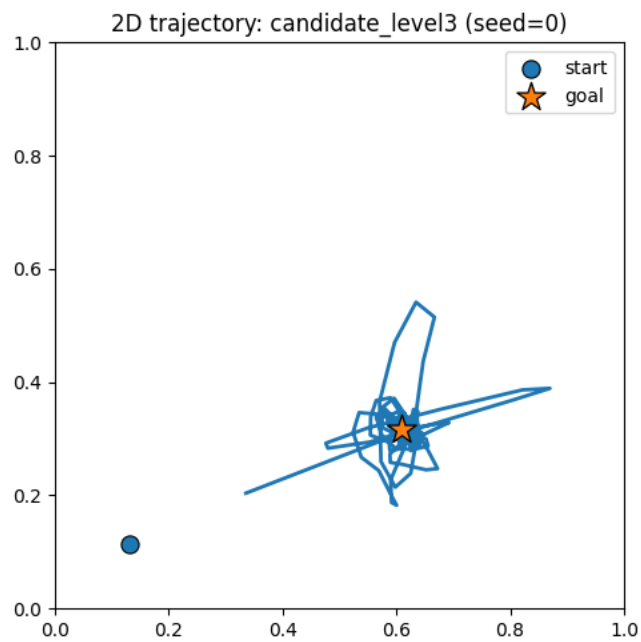


Figure 5. 2D trajectory example for the Level-3 candidate showing start, goal, and path under hidden goal + disruption.

11.6 Caveat: belief reset affects efficiency but not success

One important limitation is that Ablation C (belief reset) does not collapse post-resume success in this regime: *candidate_ablate_belief_reset* remains at 1.000 post-resume success, although resumption latency worsens (mean 12.62 vs 4.44 for the intact candidate).

Interpretation:

- This ablation presently functions as a stress test of efficiency, not a necessity proof for belief persistence.
- The result suggests that the current observation structure still allows sufficient re-localisation after resume (e.g., wall sensors returning) such that a wiped belief can be rebuilt quickly enough to preserve success.

A stricter necessity test for belief dynamics would require strengthening the evaluation (e.g., sustained partial observability post-resume, repeated belief resets, or permanently removing localisation channels), such that belief persistence becomes structurally required rather than merely helpful.

11.7 Supported attribution and epistemic scope

Supported attribution (Level-3, operational): *The tested candidate maintains a causally necessary internal target representation that sustains goal-conditioned policy coherence across delay and interference, and this cannot be reduced to reactive control.* This is supported by (i) strong separation from reactive/default baselines under post-resume success, and (ii) target-pathway ablations producing decisive collapse toward baseline performance.

Not supported (explicitly out of scope): These results do not imply agency, intrinsic motivation, selfhood, autobiographical memory, temporal identity, or subjective experience. They are evidence of a constrained organisational building block only, consistent with the program's stated epistemic restraint.

12. Conclusion

This paper defined and evaluated Level-3: Goal-Directed Policy Persistence within the staged Conscious Machines research programme. Level-3 requires that a system maintain behaviour oriented toward an internally represented target across goal omission, delay/partial observability, and interference, and that this persistence be causally dependent on the internal target pathway rather than reducible to reactive stimulus control.

Under the validity-adjusted evaluation (success counted only post-resume, i.e., after delay and interference end), the reported outcomes support a Level-3-positive interpretation under the operational definition used here.

Across multi-run trials, the candidate system consistently showed:

- High post-resume success under hidden goal, delay masking, and interference.
- Fast resumption toward the originally cued target after disruption.
- Non-reducibility to reactive or pseudo-goal baselines (reactive, default-attractor, heading-script).
- Causal necessity of the goal pathway: scrambling the goal register or decoupling the goal input from policy collapses performance toward baseline levels.

This result should be interpreted as infrastructure validation: a minimal, reproducible instantiation of Level-3 mechanisms with causal tests, rather than evidence that goal persistence emerges spontaneously under training.

Importantly, this is a bounded conclusion in the same spirit as Level-2: even strong Level-3 performance does *not* imply agency, selfhood, temporal identity, or subjective experience. It supports only the narrow claim that the tested system demonstrates goal-conditioned persistence via an internal target representation that is causally relevant to action selection across delay and interference.

13. Next step — Level-4: Temporal Self-Continuity

Level-3 establishes a minimally testable form of goal persistence under omission and interference. The next evolution is Level-4, which adds a new requirement: the system must maintain and use a causally relevant representation of its own history/continuity across time, not just a currently active goal register. Operationally, Level-4 should reuse the Level-3 structure (cue → omission → interference) but introduce multi-phase or multi-episode tasks where the correct policy depends on *what happened earlier* (deferred commitments, identity-bound constraints, or cross-context retrieval), and where Level-3 goal persistence alone is insufficient.

The Level-4 claim must remain falsifiable via continuity-state interventions (reset/scramble/swap the continuity representation; decouple it from the policy) and must be validated against strong baselines including the intact Level-3 system. Level-4 is supported only if history-dependent behaviour reliably persists across disruption and collapses under continuity-pathway ablations—constituting infrastructure validation rather than any claim of agency or subjective experience.